

CHAPTER 2: LITERATURE REVIEW AND PROJECT METHODOLOGY

2.1 Introduction

This chapter provides a comprehensive overview of the existing research and the methodological foundation for developing an AI-Powered mobile application for food recognition, halal status detection, and recipe retrieval. The literature review covers three primary areas: (1) the current landscape of food recognition technologies and their application in mobile platforms; (2) the significance of halal food compliance for Muslim consumers and the role of digital tools in supporting dietary decision-making; and (3) the application of deep learning and computer vision, particularly YOLOv8, in object detection tasks relevant to this project. The chapter also establishes the research methodology adopted for FOODIE-HALAL-SNAP, grounded in a user-centered design (UCD) approach to ensure the resulting application is technically sound, culturally relevant, and practically usable.

2.2 Facts and Findings

2.2.1 Domain

FOODIE-HALAL-SNAP operates at the intersection of three domains: computer vision and deep learning for visual food recognition, halal food compliance in Malaysia, and mobile application development for AI-powered tools. Understanding each domain is essential to establishing the context and technical direction of this project.

Computer vision is the field of artificial intelligence concerned with enabling machines to interpret and understand visual information from images and video. In the context of food recognition, computer vision techniques are applied to automatically identify food items from photographs, a task that involves significant challenges due to the high visual variability of food presentation across different cooking styles, lighting conditions, and camera angles. Deep learning, particularly Convolutional Neural Networks (CNN), has become the dominant approach for visual food recognition because of its ability to learn hierarchical feature representations directly from raw pixel data without manual feature engineering (Krizhevsky et al., 2012). Object detection, a sub-task of computer vision, extends image classification by also localising objects within the image using bounding boxes, enabling identification even when the image contains background clutter or partial occlusion.

Halal food compliance is governed in Malaysia by the Department of Islamic Development Malaysia (JAKIM) under the Malaysian Standard MS 1500:2019 (Department of Standards Malaysia, 2019). Under this standard, food is classified into three categories: halal, meaning permissible under Islamic law; haram, meaning forbidden, including pork derivatives, alcohol, and meat from animals not slaughtered according to Islamic rites; and syubhah, meaning doubtful or uncertain halal status. For Muslim consumers, verifying the halal status of food is a religious obligation, and the inability to do so when encountering an unfamiliar prepared dish poses a real daily challenge.

Mobile application development for AI tools has advanced significantly with the introduction of lightweight mobile inference runtimes such as TensorFlow Lite (TFLite) and cross-platform frameworks such as Flutter. These technologies enable deep learning models trained on powerful desktop hardware to be deployed and executed directly on consumer-grade mobile devices without requiring cloud connectivity, making AI-powered features practical for everyday mobile applications (Google, 2023).

In addition to computer vision and mobile development, this project also incorporates the use of web-based Application Programming Interfaces (APIs) for dynamic data retrieval. Specifically, the recipe retrieval functionality integrates with TheMealDB API, an open-source online database that provides structured recipe information including ingredients, cooking instructions, and meal categories. The use

of an external API enables the system to access a broader and continuously updated set of recipes without requiring large local storage, improving scalability and maintainability of the application.

2.2.2 Existing System

A review of existing systems related to food recognition and halal verification was conducted to identify the current state of available solutions, their capabilities, and the gaps that motivate the development of FOODIE-HALAL-SNAP. Five systems were reviewed and are discussed below.

The first system reviewed is **eHalal: Halal Scanner & Map**, developed by EHALAL CO., LTD. (2026). This mobile application enables users to scan the barcode of a packaged food product and retrieve its halal certification status from a JAKIM-linked database. It also provides a map feature to locate nearby halal-certified restaurants. While eHalal is effective for verifying labelled packaged food products, it is entirely dependent on the presence of a barcode, making it inapplicable to unpacked prepared dishes encountered at restaurants or social gatherings, which is the primary use case of FOODIE-HALAL-SNAP.

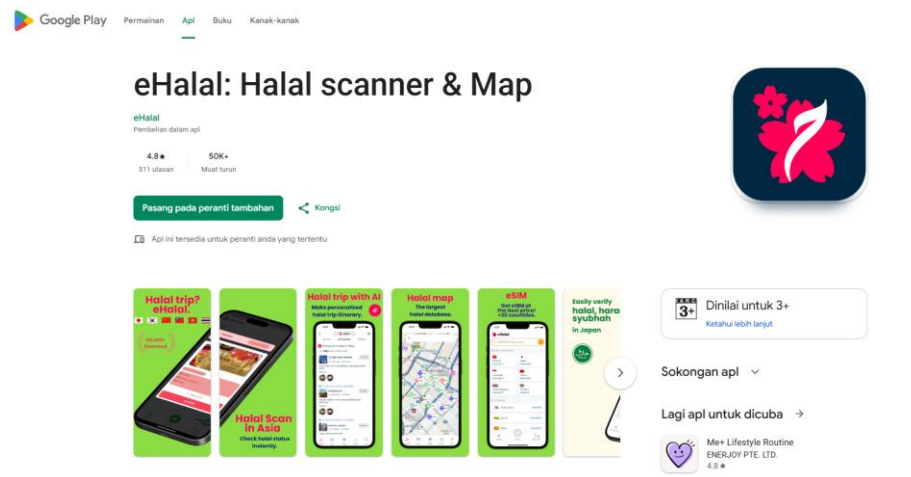


Figure 2.1: eHalal: Halal scanner & Map on Google Play

The second system is **Food For You: Food Scanner**, developed by Pastukhov (2026). This AI-powered mobile application analyses food images to provide nutritional information, allergy detection, and additive explanations. The application uses computer vision to recognise food from photos, making it conceptually similar to FOODIE-HALAL-SNAP in its image-based input approach. However, Food For You is primarily designed for Western food markets and focuses on nutritional profiling rather than Islamic dietary compliance. It does not classify food according to halal, haram, or syubhah status, nor does it cover Malaysian local dishes, limiting its utility for the target users of this project.

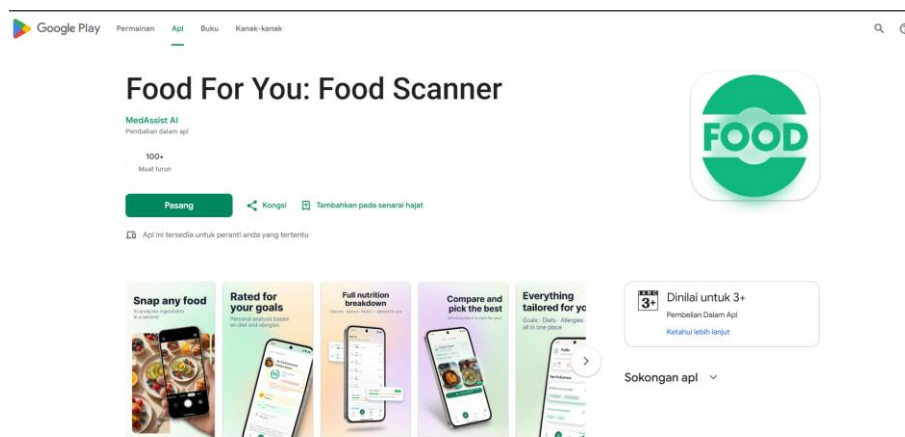


Figure 2.2: Food For You: Food Scanner on Google Play

The third system is **HALALCheck**, developed by Tarannum et al. (2024). HALALCheck is a research system that uses a multi-faceted AI approach combining image analysis and natural language processing to recognise packaged food products and classify their ingredients as halal or non-halal based on an ingredient database. This system represents the closest prior work to the halal classification module of FOODIE-HALAL-SNAP in terms of its use of AI for halal verification. However, HALALCheck targets packaged food with printed ingredient labels and does not support image-based identification of prepared dishes without ingredient labelling, which remains an unaddressed gap.

The fourth system is **iCook**, developed by Junaidy et al., Group 8 (2022) at Universiti Teknikal Malaysia Melaka. iCook is a prototype mobile application that performs image-based recognition of Malaysian food dishes. It demonstrates the technical feasibility of identifying local dishes from photographs within the Malaysian context. However, iCook does not incorporate halal status classification or recipe retrieval, limiting its practical utility for users who also need to verify dietary compliance or learn how to prepare an identified dish.

The fifth system is **iFruits**, developed N., Nordin et al, Group 13 (2022), also at Universiti Teknikal Malaysia Melaka. iFruits is a prototype application for recognising fruit items from images in a local Malaysian context. While its domain is limited to fruit categories rather than cooked dishes, it serves as a relevant reference for on-device image classification implementation in a Malaysian academic project context. Like iCook, it does not address halal compliance or recipe retrieval.

Based on the review of these five systems, it is evident that no existing application integrates all three core functionalities required by FOODIE-HALAL-SNAP: image-based recognition of Malaysian dishes, JAKIM-compliant halal status classification, and recipe retrieval. Table 2.1 summarises the reviewed systems for comparison.

Table 2.1: Summary of Reviewed Existing Systems

No.	System / App	Developer	Approach	Strength / Limitation
1	eHalal: Halal Scanner & Map	EHALAL CO., LTD. (2026)	Barcode scanning, JAKIM database lookup	Strength: Reliable halal verification for packaged products. Limitation: Inapplicable to unpacked prepared dishes where no barcode exists.
2	Food For You: Food Scanner	Pastukhov (2026)	AI image analysis, nutritional profiling	Strength: Image-based food recognition with allergy and additive detection. Limitation: Targets Western cuisine; no halal classification or Malaysian dish support.
3	HALALCheck	Tarannum et al. (2024)	AI image analysis, NLP	Strength: Multi-faceted halal ingredient

No.	System / App	Developer	Approach	Strength / Limitation
			ingredient classification	recognition for packaged food. Limitation: Requires printed ingredient labels; not applicable to image-only prepared dish identification.
4	iCook	FTMK Bengkel 2, Group 8 (2022)	Image-based Malaysian food recognition	Strength: Demonstrated feasibility of local food identification from images. Limitation: No halal classification or recipe retrieval; limited dish coverage.
5	iFruits	FTMK Bengkel 2, Group 13 (2022)	Image-based fruit recognition	Strength: Proves viability of on-device image classification for local produce. Limitation: Limited to fruit categories with no food dish or halal support.

2.2.3 Technique

A review of techniques applicable to food recognition, halal classification, and mobile deployment was conducted. Table 2.2 summarizes the reviewed techniques. Following the table, the selected technique is identified and justified, and alternative applicable techniques are discussed with reasons for their non-selection.

Table 2.2: Summary of Reviewed Techniques

No.	Technique	Category	Key Characteristic	Applicability to This Project
1	YOLOv8	Object Detection	Single-pass anchor-free detection; TFLite export support	Best speed-accuracy balance for mobile food detection; directly exportable to TFLite for on-device inference.
2	ResNet-50	Image Classification (CNN)	Deep residual learning; strong transfer learning performance	Classification-only; no spatial localization. Evaluated as baseline; superseded by YOLOv8 for detection tasks.
3	MobileNetV3	Image Classification (CNN)	Lightweight mobile-optimized architecture with low FLOPs	Classification-only architecture lacks bounding box output; lower accuracy than YOLOv8 on food detection benchmarks.
4	Faster R-CNN	Object Detection	Two-stage region proposal and classification pipeline	High accuracy but excessive inference latency for mobile deployment; incompatible with three-second response requirement.
5	EfficientDet	Object Detection	Compound scaling for efficient detection	Competitive accuracy but higher mobile inference latency than YOLOv8 in published benchmarks (Tan et al., 2020).
6	SVM with SIFT/HOG	Traditional ML Classifier	Handcrafted feature extraction with SVM classification	Demonstrated limited accuracy on food datasets with high intra-class variability (Chen et al., 2012); superseded by deep learning.

No.	Technique	Category	Key Characteristic	Applicability to This Project
7	Rule-based JAKIM DB Lookup	Halal Classification	Deterministic lookup against JAKIM ingredient database	Guarantees 100% accuracy for known dish profiles; directly aligned with MS 1500:2019 standard; no risk of misclassification from visual inference.
8	ML Ingredient Classifier	Halal Classification	Visual feature-based halal prediction	Halal status cannot be reliably inferred from visual appearance alone; misclassification risk unacceptable for religious compliance.
9.	TheMealDB	Recipe Retrieval (API)	RESTful API providing meal data including ingredients and instructions.	Enables dynamic recipe retrieval without storing large local database and real-time data access.

Based on the technique review, YOLOv8 is selected as the food detection algorithm and the rule-based JAKIM database lookup is selected for halal classification. For mobile deployment, TensorFlow Lite is selected as the inference runtime and Flutter is selected as the application development framework.

Several other approaches are also applicable to this project but were not selected for the following reasons. ResNet-50 and MobileNetV3 are applicable alternatives for dish classification as both support transfer learning and mobile deployment. However, both are classification-only architectures without native bounding box output, making them less robust than YOLOv8 when the captured image contains background clutter, partial occlusion, or multiple objects. Since users will photograph dishes in real-world environments rather than a controlled setting, the spatial localisation capability of YOLOv8 provides a meaningful practical advantage.

Faster R-CNN is an applicable and high-accuracy detection alternative. However, its two-stage region proposal pipeline incurs significantly higher inference latency than YOLOv8 on mobile hardware, making it incompatible with the three-second response time requirement. EfficientDet is similarly applicable and achieves

competitive accuracy through compound scaling but published benchmark comparisons show higher mobile inference latency than equivalent YOLOv8 configurations (Tan et al., 2020). SVM with SIFT or HOG features is not considered applicable as it has been conclusively shown to achieve inferior accuracy on food recognition datasets with high intra-class variability compared to deep learning methods (Chen et al., 2012).

For halal classification, a machine learning ingredient classifier is an applicable alternative approach, as demonstrated by Tarannum et al. (2024) in the HALALCheck system. However, this approach is not selected because the halal status of a dish depends on the precise sourcing and composition of its ingredients, which cannot be reliably determined from visual appearance alone. A fried rice dish may appear visually identical whether cooked with halal or non-halal chicken stock, yet the correct classification differs. Given the religious significance of halal compliance for Muslim users, the risk of misclassification from a visual inference approach is unacceptable, and the deterministic rule-based JAKIM database lookup is therefore the most appropriate method for this project.

For recipe retrieval, a web-based API approach is selected instead of a fully local database. TheMealDB API provides structured meal data that can be accessed dynamically at runtime. This approach reduces application storage requirements and allows access to a wider variety of recipes compared to a static local dataset. Although this introduces dependency on internet connectivity for recipe retrieval, it significantly improves scalability and ease of future updates.

2.3 Project Methodology

The development of FOODIE-HALAL-SNAP follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, adapted for a mobile deep learning application context. CRISP-DM is selected because it provides a structured, iterative workflow that systematically addresses data understanding, modelling, and deployment, which are the core technical activities of this project. The methodology is widely adopted for AI and machine learning projects and supports iterative refinement at each phase based on evaluation findings (Shearer, 2000). Figure 2.1 illustrates the adapted CRISP-DM workflow for this project.

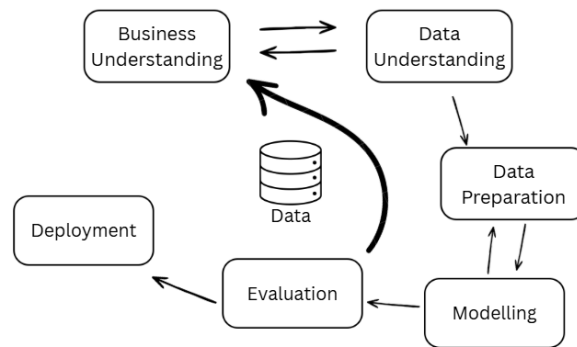


Figure 2.1: Adapted CRISP-DM Methodology Workflow for FOODIE-HALAL-SNAP

The methodology is applied across the following phases:

2.3.1 Phase 1: Business Understanding

This phase defines the project objectives and requirements from a user perspective. The three problem statements are identified: inability to identify unfamiliar Malaysian dishes from images, lack of an image-based halal verification tool for prepared dishes, and absence of a Malaysian-specific food recognition mobile application. The measurable project objectives are established, and the target user groups — Muslim consumers, home cooks, and cooking beginners — are characterized.

2.3.2 Phase 2: Data Understanding

This phase involves the collection and initial exploration of the training datasets. The Malaysia Food 11 dataset (Chan, 2020) provides 11,000 labelled images across 11 Malaysian dish categories. For this project, seven relevant categories are selected, namely nasi lemak, satay, laksa, roti canai, fried noodles, popiah, and kaya toast. Additionally, the myFood11 dataset (Mohd Subhi, n.d.) is utilised to supplement the dataset with three additional categories: nasi goreng, mee goreng, and curry puff. Data understanding activities include analyzing class distribution across categories, identifying data quality issues such as mislabeled images or extreme outliers, and assessing the adequacy of dataset size per category for model training. A minimum of 800 images per category is targeted to ensure sufficient training data for the YOLOv8 fine-tuning process.

2.3.3 Phase 3: Data Preparation

Data preparation covers all preprocessing and augmentation activities applied to the raw image datasets before model training. Images are resized to 640 x 640 pixels to match the YOLOv8 input specification and normalized to a [0, 1] pixel value range. Data augmentation techniques are applied to increase training set diversity and improve model generalization, including random horizontal flipping, random brightness, and contrast adjustment with factors in the range [0.8, 1.2], and mosaic augmentation, which combines four training images into a single composite image. The dataset is split into training (70%), validation (20%), and test (10%) subsets using stratified sampling to preserve class distribution across splits. Images with severe occlusion, extreme blur, or ambiguous content are removed during a manual inspection pass.

2.3.4 Phase 4: Modelling

The modelling phase covers the training and optimization of the YOLOv8n food detection model. Transfer learning is applied by initializing the model with YOLOv8n weights pre-trained on the COCO dataset, which are then fine-tuned on the prepared Malaysian food image dataset. Training is conducted for a maximum of 100 epochs with early stopping based on validation mAP@0.5, using the Adam optimizer with an initial learning rate of 0.001 and a cosine annealing learning rate schedule. The batch size is set to 16.

2.3.5 Phase 5: Evaluation

The evaluation phase assesses the trained model and the complete application against the performance metrics defined in Chapter 3 (Section 3.3.3). Model performance is evaluated on the held-out test set using mAP@0.5, precision, and recall per dish category. End-to-end application response time is measured on target mobile hardware. Halal classification accuracy and application usability are assessed as described in the requirement analysis. If evaluation results do not meet the acceptance thresholds, the methodology supports iteration back to the Data Preparation or Modelling phase for refinement.

2.3.6 Phase 6: Deployment

The deployment phase covers the packaging and distribution of the completed Flutter application for Android platforms. The Android build is generated as an APK file. The application is tested on representative physical devices before release. Post-deployment maintenance activities include monitoring for reported defects, updating the halal ingredient database to reflect any revisions to JAKIM guidelines, and extending the dish detection model to additional Malaysian categories in future iterations.

2.4 Project Requirements

This section describes the software, hardware, and other requirements for the development and deployment of FOODIE-HALAL-SNAP.

2.4.1 Software Requirement

- Python 3.10 or above — model training, dataset pre-processing and TFLite model conversion scripts.
- Ultralytics YOLOv8 library (ultralytics $\geq$ 8.0) — YOLOv8 model training, validation and export.
- PyTorch $\geq$ 2.0 with CUDA support — deep learning framework for model training with GPU acceleration.
- TensorFlow Lite (TFLite) runtime — on-device mobile inference for the deployed food detection model.
- Flutter SDK $\geq$ 3.0 — cross-platform mobile application development for Android.
- Dart SDK (bundled with Flutter) — programming language for Flutter application development.
- tflite_flutter plugin — Flutter plugin for integrating TFLite model inference within the application.
- Android Studio — Build tools for Android application compilation and testing.
- LabelImg or Roboflow — image annotation tool for bounding box labelling of training dataset images.
- OpenCV (Python) — image pre-processing and augmentation during dataset preparation.
- HTTP/REST API Integration (TheMealDB API) – Used for retrieving recipe data dynamically from an external online database.

2.4.2 Hardware Requirement

- Development PC: Minimum Intel Core i7 (or equivalent AMD Ryzen 7), 16 GB RAM, NVIDIA GPU with minimum 6 GB VRAM (e.g. NVIDIA RTX 3060) for YOLOv8 model training.

- Storage: Minimum 50 GB available disk space for datasets, model files, and development tools.
- Target Android test device: Minimum Snapdragon 665 equivalent processor, 4 GB RAM, Android 8.0 (API 26) or above, rear camera.
- Webcam / device camera: Used for live image capture testing during development.

2.4.3 Other Requirement

- Internet connection: Required during development for dataset downloading, package installation, and cloud-based training monitoring. Not required by the deployed application for core functionality.
- JAKIM halal classification reference materials: Access to MS 1500:2019 Malaysian Standard for Halal Food and JAKIM published ingredient permissibility guidelines for construction of the halal classification database.
- TheMealDB API access: Internet access to retrieve recipe data dynamically from TheMealDB online database.

2.5 Project Schedule and Milestones

The project activities are planned across two semesters, PSM 1 and PSM 2. The activities are structured stage by stage in accordance with the CRISP-DM methodology phases, ensuring each phase is completed and validated before the subsequent phase begins. Table 2.4 lists the activities stage by stage with their associated milestones and deliverables.

Table 2.3: Project Schedule and Milestones

No.	Activity	CRISP-DM Phase	Semester	Milestone / Deliverable
1	Problem identification and domain review	Business Understanding	PSM 1	Documented problem statements and project objectives.
2	Literature review: existing systems and techniques	Business Understanding	PSM 1	Completed Chapter 2 literature review with 20+ references.
3	Dataset collection: Malaysia Food 11 and myFood11	Data Understanding	PSM 1	Downloaded and verified datasets; class distribution report.
4	Dataset preprocessing and augmentation	Data Preparation	PSM 1	Completed preprocessed and augmented dataset; train/val/test splits.
5	System design: architecture, database schema, UI wireframes	Data Preparation / Modelling	PSM 1	Completed system architecture diagram, database schema, UI wireframes.
6	PSM 1 report writing and submission	—	PSM 1	Submitted PSM 1 report covering Chapters 1–3 and Chapters 4 for draft.
7	YOLOv8 model training and evaluation	Modelling	PSM 2	Trained model achieving minimum mAP@0.5 of 80%; exported to TFLite.
8	Halal classification database development	Modelling	PSM 2	Completed JAKIM-aligned halal status

No.	Activity	CRISP-DM Phase	Semester	Milestone / Deliverable
				database for all ten dish categories.
9	Recipe database development	Modelling	PSM 2	Completed local recipe database with ingredients and steps for all ten categories.
10	Flutter mobile application development and integration	Deployment	PSM 2	Completed functional Android application with all modules integrated.
11	System testing: functional, performance and usability	Evaluation	PSM 2	Completed evaluation report against all defined performance metrics.
12	PSM 2 final report writing and submission	—	PSM 2	Submitted complete final report covering all chapters.

2.6 Summary

This chapter has presented the literature review and project methodology for FOODIE-HALAL-SNAP. Section 2.2 identified the three relevant domains — computer vision and deep learning, halal food compliance, and mobile application development — and reviewed existing systems and techniques. The review of existing systems revealed that no current application integrates image-based Malaysian food recognition, JAKIM-compliant halal classification, and recipe retrieval via TheMealDB API. The technique review confirmed YOLOv8 as the optimal detection architecture, the rule-based JAKIM database as the most reliable halal classification approach, and Flutter with TFLite as the most suitable mobile development stack. Section 2.3 described the adapted CRISP-DM methodology adopted for this project, covering all six phases from business understanding through to deployment, with defined experiment parameters in Table 2.3. Section 2.4 listed the software, hardware, and other project requirements.